



Report on Business Models for Solar Projects in Assam

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Asian Development Bank

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List of Abbreviations

AEGCL: Assam Electricity Grid Corporation Limited
AERC: Assam Electricity Regulatory Commission
ALMM: Approved List of Models and Manufacturers
APDCL: Assam Power Distribution Company Limited
APGCL: Assam Power Generation Corporation Limited
BESS: Battery Energy Storage System
BOO: Build, Own, Operate
BOOT: Build, Own, Operate, Transfer
CAGR: Compound Annual Growth Rate
CEA: Central Electricity Authority
COD: Commercial Operation Date
CPSU: Central Public Sector Undertaking
DISCOM: Distribution Company
DRE: Distributed Renewable Energy
DSM: Demand Side Management
EPC: Engineering, Procurement, and Construction
ESO: Energy Storage Obligation
ESS: Energy Storage System
FSP: Floating Solar Project
FY: Financial Year (In the current context it is 1-April till 31-March)
GDAM: Green Day-Ahead Market
GTAM: Green Term-Ahead Market
GW: Gigawatt
GWh: Gigawatt hours
HPO: Hydro Purchase Obligation
IPP: Independent Power Producer
ISTS: Interstate Transmission System
KSEB: Kerala State Electricity Board
KW: Kilowatt
MMBT: Million Metric British Thermal Units
MOU: Memorandum of Understanding
MVA: Mega Volt-Ampere
MW: Megawatt
OEM: Original Equipment Manufacturer
PPA: Power Purchase Agreement
PPP: Public-Private Partnership
PSP: Pumped Storage Plant
REC: Renewable Energy Certificate
RES: Renewable Energy Source
RESCO: Renewable Energy Service Company
RPO: Renewable Purchase Obligation

SLDC: State Load Dispatch Center

SPPD: Solar Power Park Developer

TBCB: Tariff-Based Competitive Bidding

USD: United States Dollar

WPO: Windpower Purchase Obligation

2 Introduction

2.1 Background

The State of Assam has witnessed an economic growth with Compounded Annual Growth Rate (CAGR) of 11.2% from Financial Year FY12¹ to FY22² surpassing the national average of 10.7%³. The state is the gateway to the rest of the entire North-East Region in India, comprising of the other states of Arunachal Pradesh, Tripura, Meghalaya, Manipur, Mizoram and Nagaland, acting as a major economic and transport corridor to the region. Assam's strategic location near neighboring countries like Bangladesh, Bhutan, Nepal, China, and Myanmar also presents unique opportunities for economic, strategic, and cultural collaborations in the vast Asia-Pacific region.

The state's demand for electricity has also steadily increased at a CAGR of around 5.9% during FY12 to FY22, exceeding national average of 3.9% backed by the economic development, increasing standard of living and rising heat waves in the region. While the State has been able to meet most of the rising energy demand, it continues to face energy security challenges. Historically, the State has relied on gas and hydro-based power generation, facing challenges related to gas availability and price fluctuations. Further the State is dependent on electricity supply from outside the State (due to limited intra State supply capacity) for meeting the rising electricity demand. The electricity procurement costs in the State have risen sharply in recent years due to increase in fuel prices which have put significant strain on the financials of Assam Power Distribution Company Limited as it continues to cater to the entire State's electricity supply.

2.2 Need for Solar Capacity in Assam

The rising electricity demand in Assam reached 11,465 Giga Watt-Hours (GWh) with a peak electricity demand of 2,379 Mega Watts in FY23. The State's electricity supply capacity which historically was in deficit has reached parity with demand in FY23. In terms of supply capacity, the State has an installed capacity of 1,993 MW of which thermal sources comprise ~68% share followed by Hydro (18%), Solar (12%) and other Renewable Energy Sources at 2% in FY23. The share of Solar energy has increased from 41 MW in FY19 to 248 MW in FY23 due to push to build more clean energy capacity in Assam by the Government. However, the State continues to rely on power trading platforms to meet the electricity demand which reached a peak of ~2,500 MW in FY24 (till August 2023).

In terms of the power purchase mix of APDCL (from all sources) in FY24 (which stood at 12,886 GWh), thermal (coal and gas based sources) provided 68% of the electricity supply, followed by hydro at 17%, renewable energy including solar at 4% and remaining by power trading at 11%. The rising cost of fuel for coal and gas plants along with high cost of power in Indian power exchanges has led to a sharp increase in average power procurement costs for APDCL from Rs 4.19 / Kilo Watt-Hour (KWh) in FY22 to Rs. 5.38 / kWh in FY23. The average power procurement costs from power trading increased from Rs. 3.93 in FY21 to Rs. 7.06 / KWh in FY23.

As per the Central Government estimates, Assam has a Renewable Energy Potential of ~14 GW of which ~13.8 GW is from solar followed by small hydro at 202 MW and Bio-Power at 279 MW. With limited scope to increase gas / coal-based generation due to environmental/social considerations, gas production expansion challenge and difficulty in setting up coal supply chain infrastructure, the State has to develop its Renewable Energy Potential to meet rising electricity demand and bolster its clean energy portfolio. The Central Government has also come up with Renewable Purchase Obligation (RPO) which mandates distribution companies like APDCL to meet a percentage of their energy supply through renewable energy sources. This guideline which is revised periodically is generally adopted by respective State Electricity Regulatory

¹ Note: The Financial Year in India begins from 1st April and end on 31st March i.e. FY12 would refer to the period beginning from 1st April 2011 and end on 31st March 2012.

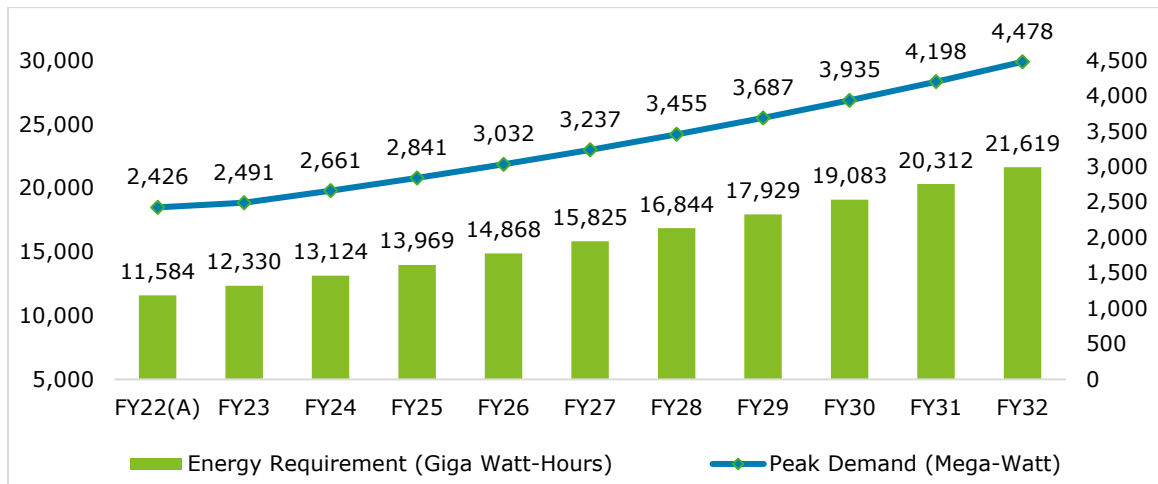
² Based on the growth of Net State Domestic Product (NSDP)

³ Source: Ministry of Statistics and Program Implementation, Government of India (MOSPI)

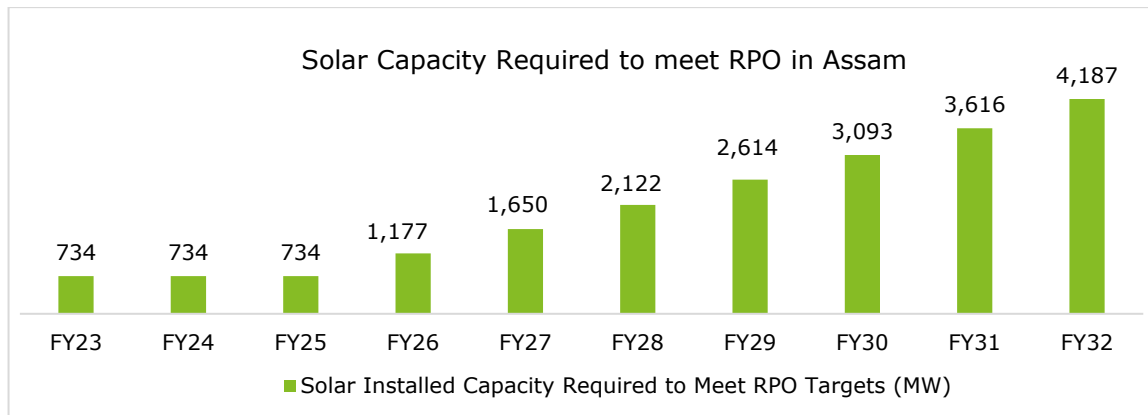
Commissions for each State. The Assam Electricity Regulatory Commission (AERC) had adopted the RPO targets set by Central Government for APDCL. As per the latest guidelines issued in October 2023, Discoms will have to procure ~43% of the electricity supply capacity from renewable energy sources such as Hydro (~1%), Wind (~3%), Distributed Energy Resources (2%) and Other RPO (36%) by FY30. The Distributed Energy Resources (DER) and Other RPO can be met through Solar capacity. This RPO target may expand to ~47% by FY32 if the government continues this increasing target trajectory beyond FY30.

In terms of electricity demand forecasts, The National Electricity Plan⁴ formulated by the Central Planning Agency for Electricity (Central Electricity Authority-CEA) in India has projected that the electricity demand in Assam is expected to nearly double from current levels to 21,619 GWh by the Financial Year (FY) 2032⁵ from current levels. The electricity demand is being driven by the economic development of the region which is helping growth across major electricity consumer categories such as domestic, commercial and irrigation among others with a Compound Annual Growth Rate (CAGR) of 7-8% for each respective category.

Figure 1: Base corrected demand projections in line with CEA's considered rate of growth⁶



Considering the planned capacity additions for Wind and Hydro projects in the State till FY30, the State would require to setup about ~4.2 GW of solar capacity by FY32 to meet the RPO targets. The YoY targets are showcased below:



⁴ The National Electricity Plan along with forecast for demand in each State of India is prepared by the Central Electricity Authority and is called the Electric Power Survey (EPS). The last published EPS report is the 20th EPS published in November 2022. The EPS is aligned to the Long Term Generation Plan of the country which is also published by CEA.

⁵ The Indian Financial Year follows an April to March period. For e.g., Financial Year 2032 (FY32) implies the period between 1st April 2031 to 31st March 2032.

⁶ Source: CEA, Deloitte Analysis; Note: Energy Requirement has been base corrected basis actual achieved in FY22 and escalated based on Year on Year growth projection rates in 20th EPS for Assam.

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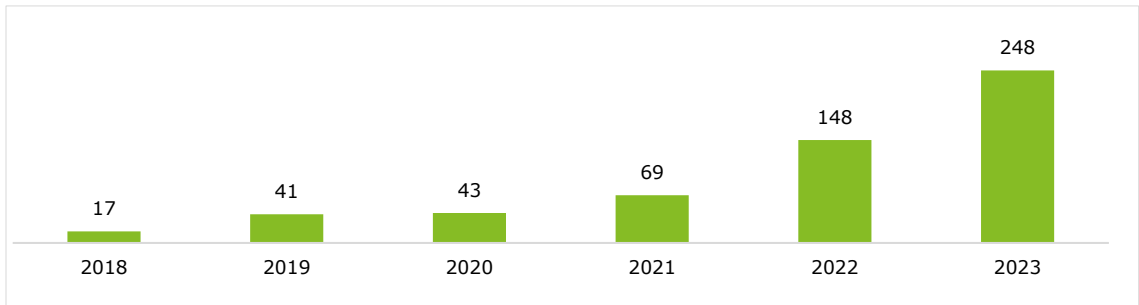
The current installed capacity in Assam is only 248 MW for solar which will require the state to add ~3.9 GW by FY32 which requires a relook at the business models that can be leveraged to build this capacity rapidly. The Government of Assam is also committed to promoting renewable energy, introducing initiatives like the Assam Renewable Energy (RE) Policy in 2022 with a target of adding 1,200 Mega-Watt (MW) of renewable energy capacity. Additionally, schemes like "Mukhya Mantri Suro Shakti Prokolpo" aim to install 1000 MW of solar projects on government-owned land, signaling a strong shift towards a greener energy portfolio.

2.3 Project Development Experience in Assam and Key Learnings

2.3.1 Current Status

In the last 5 years, Assam has added ~231 MW of solar capacity increasing the installed base by ~5X times. Most of the projects have been small-scale IPP projects with the largest single location plant having a capacity of 70 MW at Amguri. The discovered tariff for these projects has been in the range of INR 3.2-4 per kWh.

Figure 2: YoY Solar Installed Capacity Trend (MW)



In terms of the project locations, the concentration has been largely within Central Assam where most of the demand centers also currently fall. The location wise small scale solar projects have been illustrated below:

Figure 3: Solar utility scale projects located in Assam

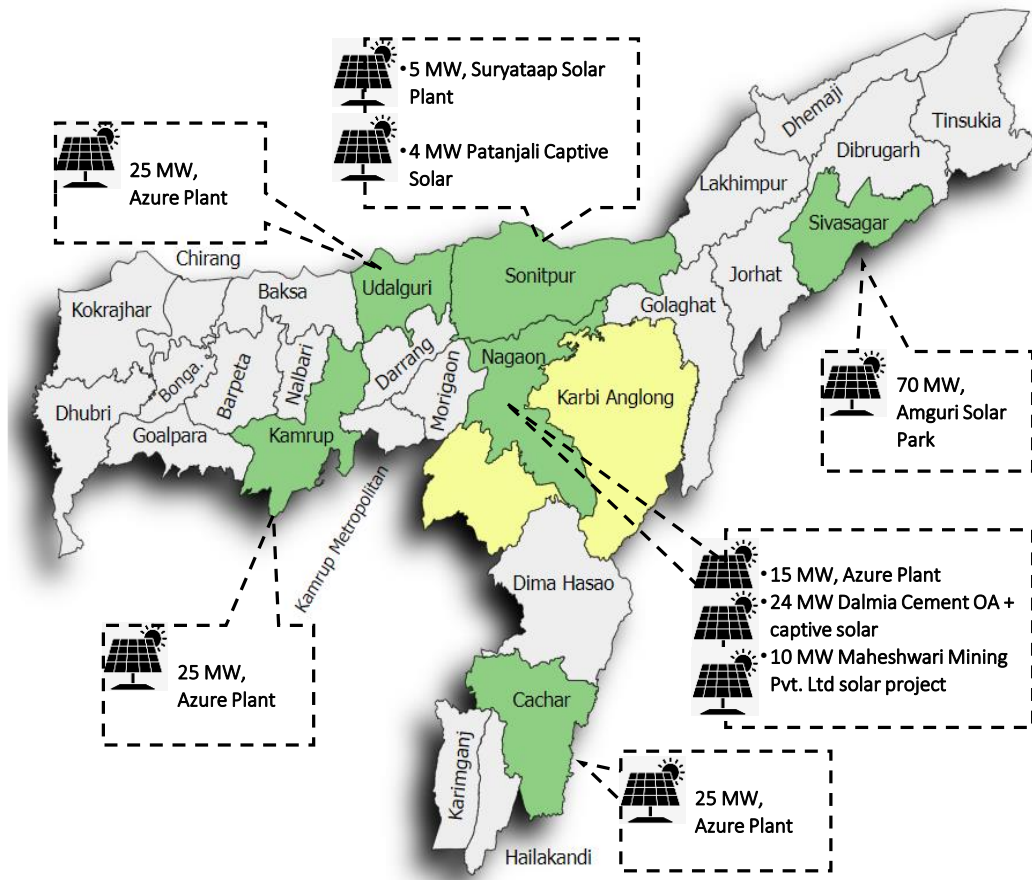


Table 1: List of utility scale solar projects in Assam

S No	Implementing Agency	Contract	Capacity (MW)	COD	Model
1	Suryataap Solar	PPA	5	20-Aug-16	-
2	Azure Power (R1)	PPA	25	12-Sep-20	BOO
3	Azure Power (R2)	PPA	25	30-Dec-21	BOO
4	Azure Power (R3)	PPA	15	27-Jan-22	BOO
5	Azure Power (R4)	PPA	25	31-Mar-22	BOO
6	Maheshwari Mining & Energy	PPA	10	31-Mar-22	BOO
7	APGCL (Developer: Jakson)	Solar Park	70	31-Mar-22	BOOT
8	Patanjali	Captive	4	NA	-
9	Dalmia Cement	Captive + OA	24	NA	-

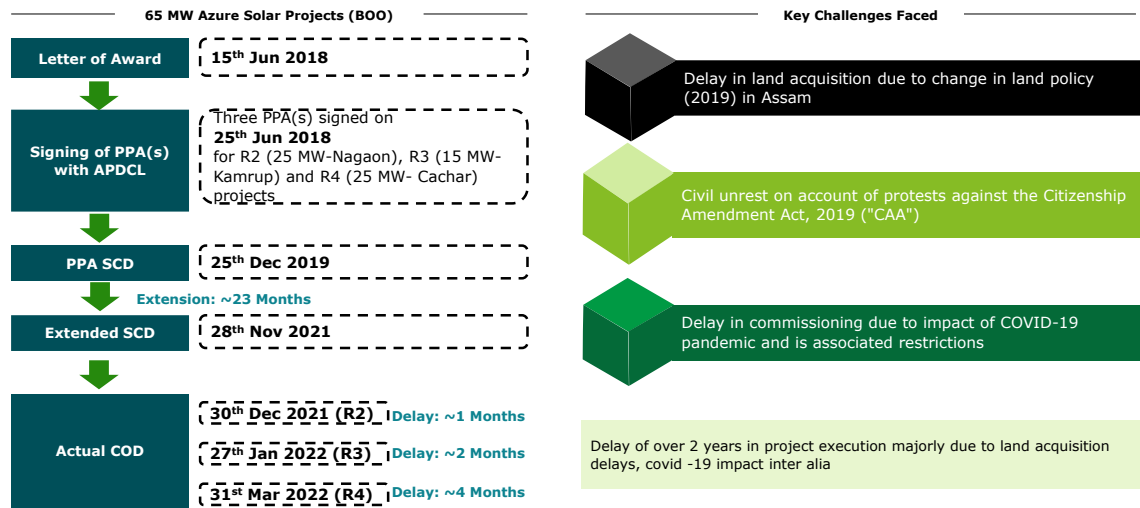
2.3.2 Key Challenges

The majority of the projects have been setup under the Build, Own and Operate (BOO) model which puts the onus of the capex and operations on the private developers. Most of these projects have taken 2.5 to 4 years in achieving commissioning from their award date due to delays pertaining to land acquisition, geographical challenges and Covid-19 related restrictions among others. The average tariffs of Rs. 3.2 – 4 / kWh is also significantly higher than the project tariffs discovered in competitive biddings conducted by Solar Energy Corporation of India (SECI) in other parts of the country where tariffs have ranged between Rs. 2.2 to 2.8 / kWh.

Some of the key projects in Assam which have and / or facing delays have been enumerated below:

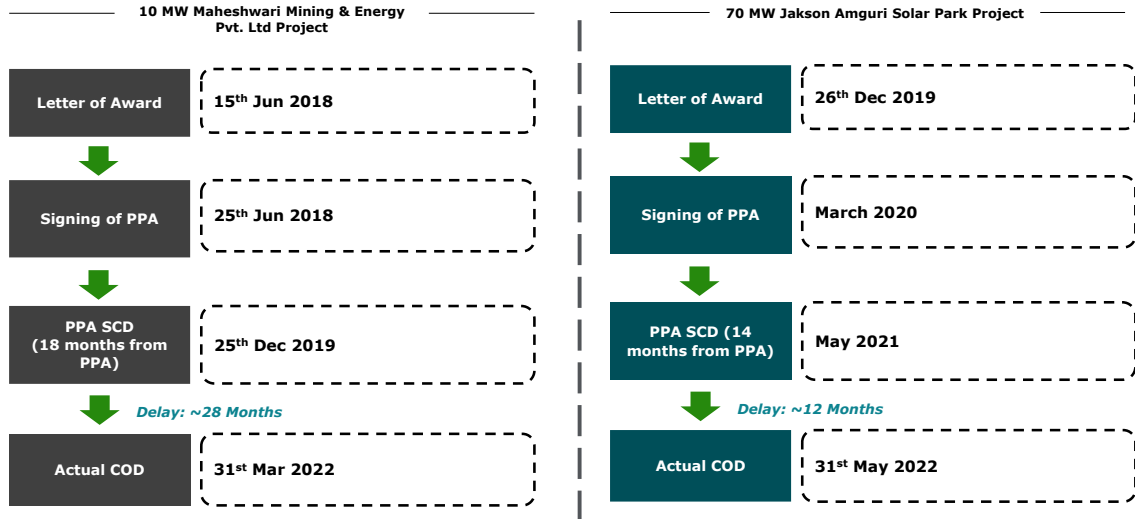
2.3.2.1 Issues faced in implementation of commissioned 65 MW Azure Power projects

Figure 4: Issues faced in implementation of commissioned 65 MW Azure Power projects



2.3.2.2 Key Timelines in project execution for 10 MW MME Pvt. Ltd and 70 MW Amguri Solar Project in Assam

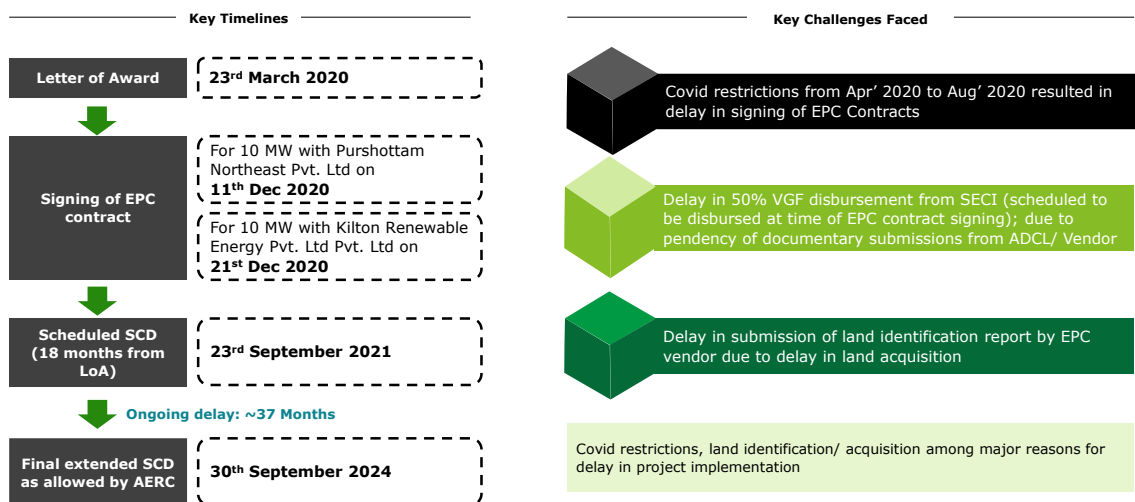
Figure 5: Key Timelines in project execution for 10 MW MME Pvt. Ltd and 70 MW Amguri Solar in Assam



2.3.2.3 Issues faced in implementation of APDCL 2 X 10 MW projects under CPSU scheme (Ongoing)

Even in case of the ongoing project development of 2x10 MW Solar project in Assam, under the Central Public Sector Unit Scheme of the Govt. of India, there has been a considerable delay due to covid, delay in Viability Gap Funding from the SECI, delay in documentary submissions and land related issues.

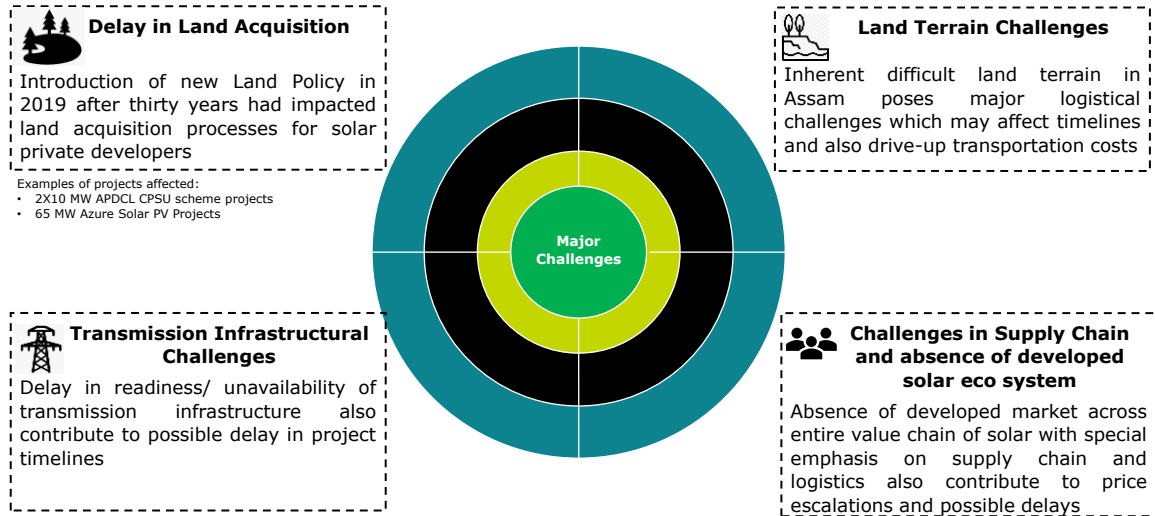
Figure 6: Issues faced in implementation of APDCL 2 X 10 MW projects under CPSU scheme



Key Conclusion:

While solar capacity has risen in Assam, the projects have historically faced multiple challenges pertaining to land acquisition, difficult terrain, power evacuation as well as solar supply chain and ecosystem. Some of these challenges have been enumerated below:

Figure 7: Key Challenges in setting up of Solar Projects in Assam



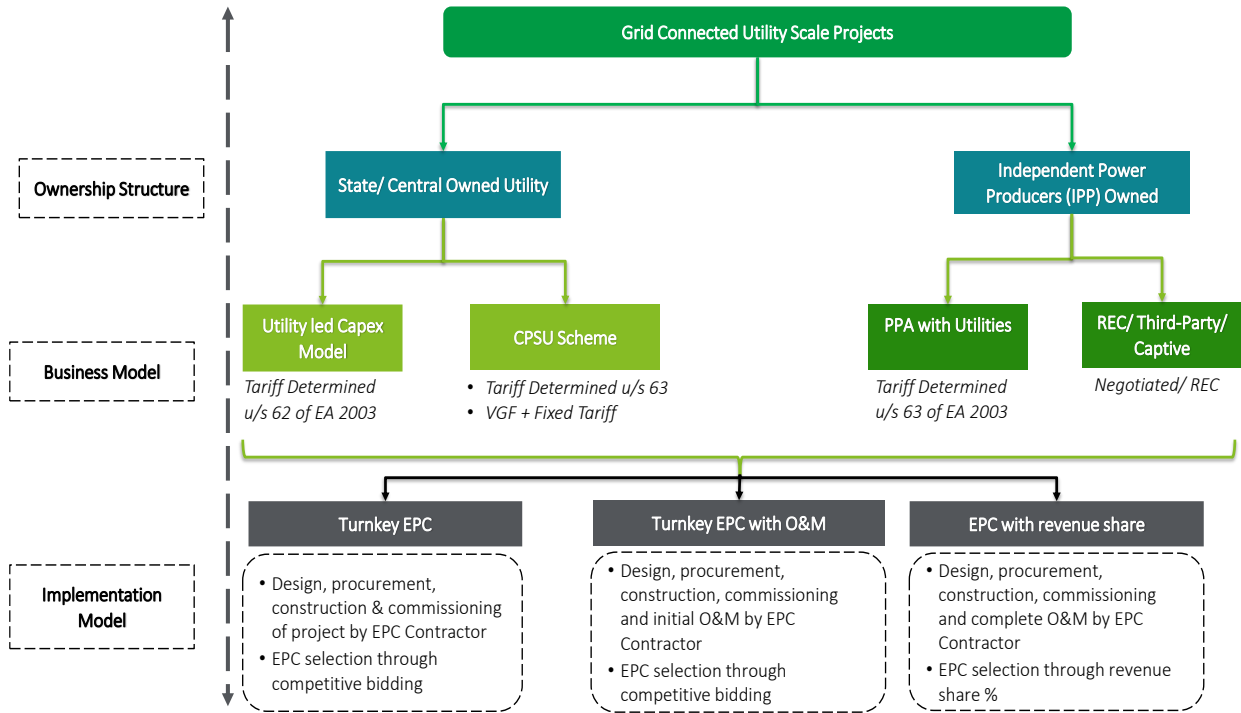
In order to scale the State's Solar Capacity to ~4200 MW by FY32, there is a need to plan mitigation techniques through policy, regulatory measures and interventions to support scalable business models. The next section discusses various business models for solar projects that are prevalent and can be adopted for Assam.

3 Business Models for Solar Projects in India and Learnings from Project Development Experience in Assam

In this section, prevalent business models for setting up Solar PV projects has been assessed which provides a foundation for developing Solar Projects in Assam going forward.

In terms of Grid Connected Utility Scale Solar Projects, 2 types of ownership-based structures are prevalent i.e. Utility Owned and Independent Power Producer (IPP) Owned. The below graph illustrates the ownership structure wise business and implementation models possible under each structure:

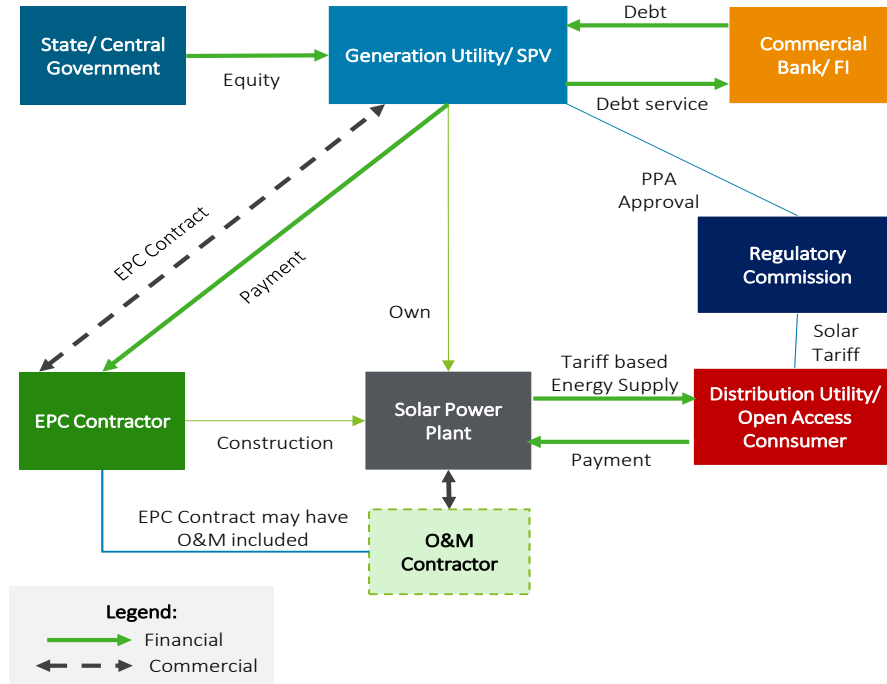
Figure 8: Business models for utility scale solar projects in India



3.1 Public Sector owned utility scale solar Model

This is a utility led model wherein capex is incurred by the utility itself; thus, ownership of such projects also lies with the utilities. Typically, in such scenarios, a Turnkey/ package-wise EPC contract is/are awarded to Vendor(s) through competitive bidding/ nomination basis.

Figure 9: Utility led Capex Model



Key Highlights of the model:

Table 2: Key Highlights of Utility Owned Capex Model

Aspect	Details
Key Features	- Government Utility builds, owns and operates the solar project - Suitable for small to large scale projects
Financing	- Equity funded by State Government/ Utility – Balance sheet based funding - Project specific Debt raised from commercial banks/ FIs
Contractual Arrangement	- Power Purchase Agreement – Where generation utility and procuring utility are different or SPV is used for solar project. No PPA is required where procuring and generating utilities are same - EPC Agreement – Between the generating utility and EPC contractor
Commercial	- Regulator to determine the tariff on cost plus (U/S 63) - Competitive selection of EPC (adoption of tariff U/S 62)
Pros and Cons	Benefits - Less complexity in transaction structuring - Builds expertise of Utility for future projects - Greater control in cost optimization Challenges - Utility's ability to develop solar projects and bring efficiencies - Limited experience in operating and managing solar projects

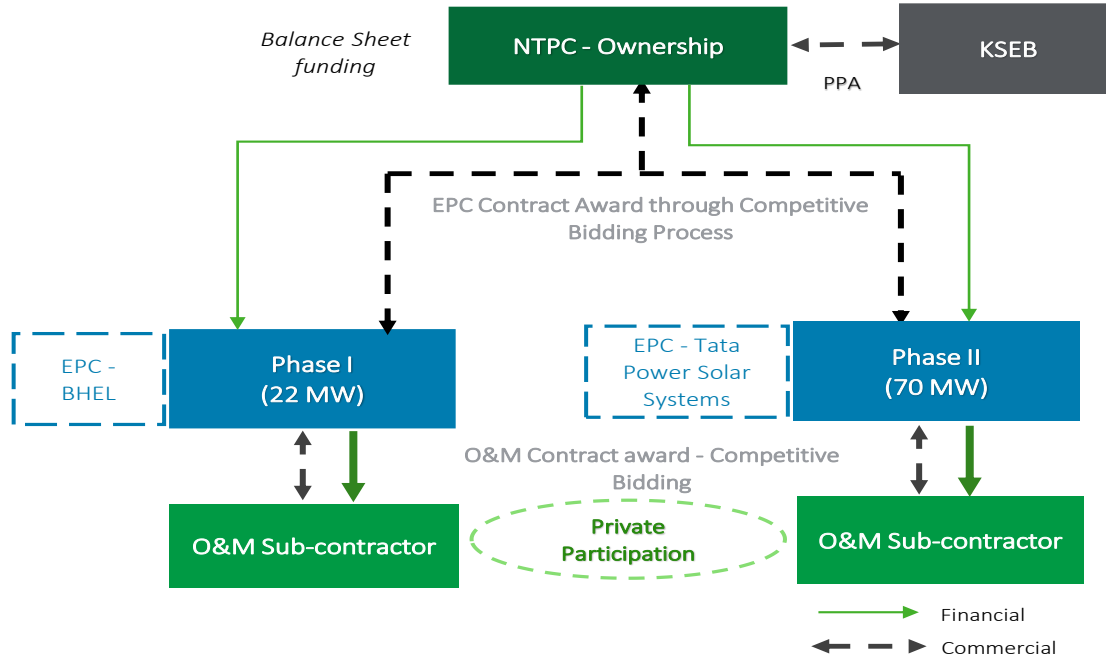
In terms of the utility model, 2 case studies of NTPC owned projects have been enumerated below:

3.1.1 Case Study: NTPC 92 MW Kayalkulam Floating Solar Plant (FSP), Kerala⁷

Project Details:

- MOU signed between the NTPC and the Kerala State Electricity Board (KSEB) in May 2018.
- NTPC initiated the bidding process for selection of EPC Contracts in two phases
- EPC cum O&M contracts awarded to two different companies
- Plant fully operational from June 2022

Figure 10: NTPC FSP (92 MW) at Kayalkulam



Key Highlights:

Table 3: Key Highlights of NTPC Kayakulam FSP Project

Aspect	Details
Project Capacity	• Phase I: 22 MW (EPC Contractor: M/S BHEL) • Phase II: 70 MW (EPC Contractor: M/S Tata Power Solar Systems Limited)
Capital Cost	Cost Approved by CERC • Phase I (22 MW): ~Rs 115 Cr • Phase II (70 MW): ~Rs 404 Cr
Funding	• Balance Sheet Financing by NTPC • D/E ratio of 80:20 as considered by CERC
Commercial	• PPA with KSEB at Rs 2.94 per unit • Tariff determined u/s 62 of EA 2003

3.1.2 Case Study: O&M and EPC contract NTPC 25 MW Simhadri Floating Solar Plant, Andhra Pradesh⁸

Project Details:

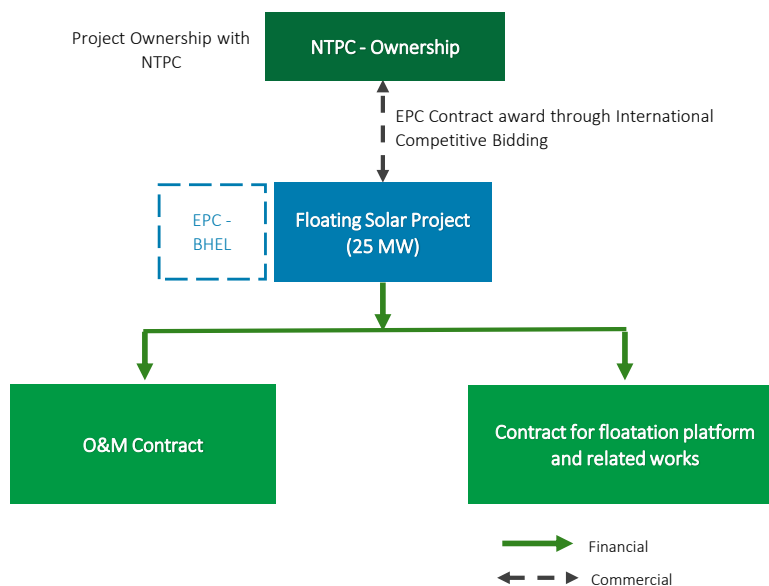
⁷ Source: Deloitte Research

⁸ Source: Deloitte Research

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- Project executed by NTPC under Flexibility in Generation and Scheduling of TPS to reduce emissions scheme of MoP(05-Apr-2018) to replace the thermal power supplied from Simhadri STPS to its beneficiaries
- Commissioning in two phases: 10 MW on 30-Jun-2021 and 15 MW on 21-Aug-2021
- Net gain from the supply of solar power in place of thermal power shared with the beneficiaries in the ratio of 50 (Beneficiary): 50 (NTPC)

Figure 11: NTPC Simhadri Floating solar project framework



Key Highlights:

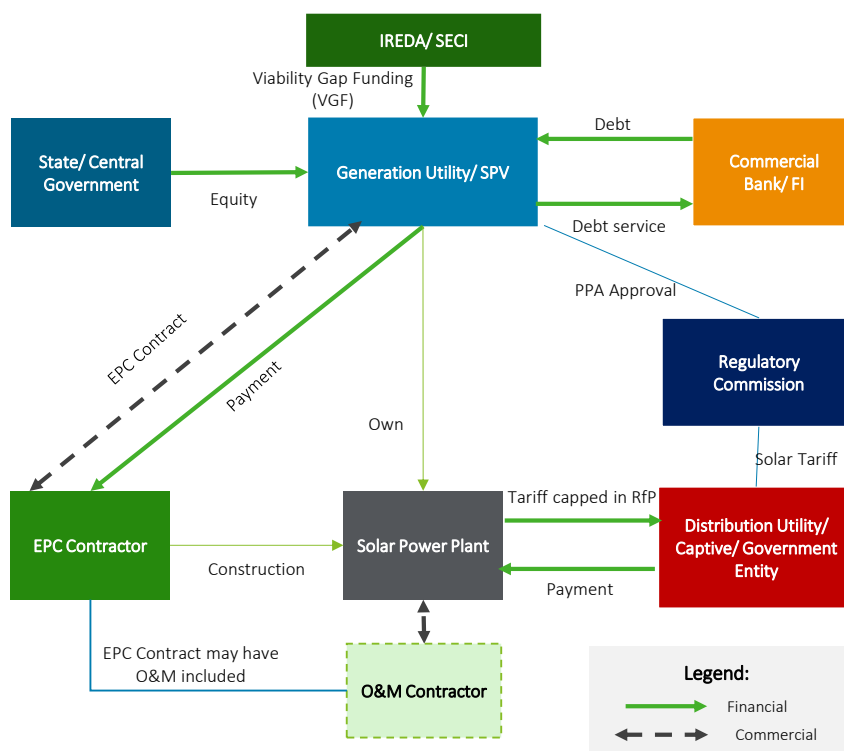
Table 4: Key Highlights of NTPC Simhadri FSP Project

Aspect	Details
Project Capacity	• 25 MW (EPC Contractor: M/S BHEL)
Capital Cost	• Cost Approved by CERC • INR 100 Crore (\$ 14.57 Mn)
Commercial	• No separate agreement and replacement of power to be considered within the existing PPA • Provision for sale of power (at ECR of Simhadri STPS) in open exchange • Landed price of RE power < Energy Charges of generating station whose power is replaced

3.2 CPSU Scheme for solar projects

This scheme gives opportunity for Central and State PSUs (CPSU) to execute solar projects with Viability Gap Funding (VGF). Under CPSU scheme, Government/ Government entities can participate in tenders wherein tariff would be capped, and basis of bid participation would be quantum of Viability Gap Funding (VGF) quoted. Responsibility of finding buyers lies with the Bid winners. The transactional overview as well key highlights of the model have been enumerated below:

Figure 12: CPSU Model - Project Overview



Key Highlights:

Table 5: Key Highlights of the CPSU Scheme

Aspect	Details
Key Features	Ministry of New & Renewable Energy (MNRE) approved implementation of CPSU Scheme Phase-II for setting up 12,000 MW grid-connected solar projects by CPSUs/ State PSUs/ Government Organizations, with Viability Gap Funding (VGF) support
Financing	- Equity funded by State Government/ Utility – Balance sheet-based funding - Project specific Debt raised from commercial banks/ FIs
Contractual Arrangement	For self-use or use by Government/ Government entities, either directly or through Distribution Companies (DISCOMS).
Commercial	- VGF capped at Rs 0.55 Cr /MW - decided through bidding - VGF released in two tranches - 50% on EPC contract Award & balance on commissioning of full capacity - Tariff capped at Rs 2.45 per kWh
Pros and Cons	Benefits - Less complexity in transaction structuring - Builds expertise of Utility for future projects Challenges - Can lead to inefficiencies in procurement - Limited experience in operating and managing solar projects

The results of some of the tenders under CPSU Scheme have been illustrated below:

Figure 13: Results of tenders under CPSU Scheme⁹

SECI TRANCHE I (Aug 2019) - Targeted Capacity : 2 GW - Awarded Capacity: 922.4 MW - Capped Tariff: Rs 3.50 per kWh			SECI TRANCHE II (Nov 2019) - Target Capacity: 1.5 GW - Awarded Capacity: 1104.8 MW - Capped Tariff: Rs 3.50 per kWh			IREDA TRANCHE III (Sep 2021) - Target Capacity: 5 GW - Awarded Capacity: 5 GW - Capped Tariff: Rs 3.50 per kWh		
Bidder's Name	VGf Quoted (Rs L/MW)	Capacity (MW)	Bidder's Name	VGf Quoted (Rs L/MW)	Capacity (MW)	Bidder's Name	VGf Quoted (Rs L/MW)	Capacity (MW)
NHDC Limited	55.00	25	Singreni Collieries Company	68.00	81	SJVN Limited	44.72	1,000
Singreni Collieries Company	60.00	90	Indore Municipal Corporation	68.80	100	NLC India Limited	44.75	510
Assam Power Distribution (APDCL)	68.00	30	NTPC Limited	70.00	923	NHPC Limited	44.90	1,000
Delhi Metro Rail Corporation (DMRC)	69.75	3				IRCON International Ltd	44.94	500
Nalanda University	69.96	5				NTPC Limited	44.95	1,990
NTPC Limited	70.00	769						

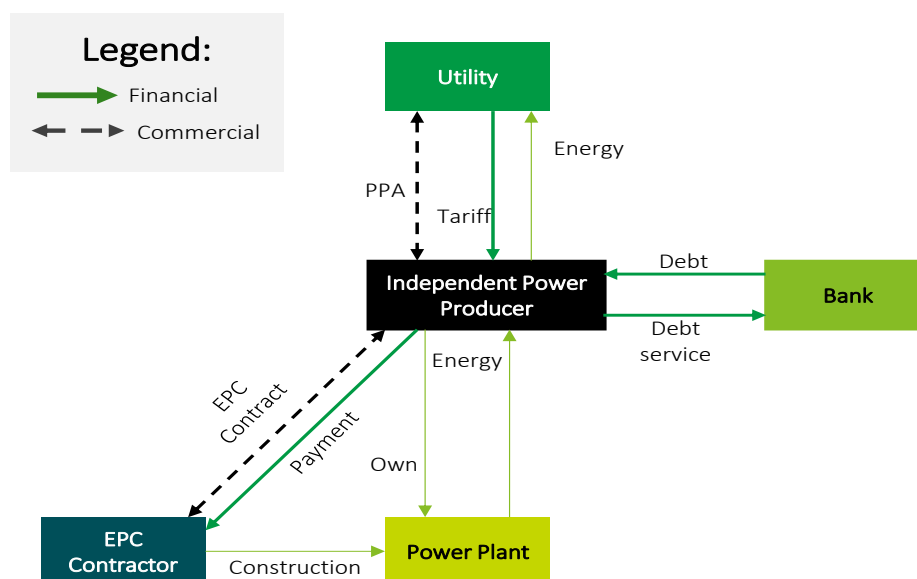
3.3 Private Owned utility scale solar models

The projects under IPP / PPP based ownership are financed by Private Independent Power Producers (IPPs) or combination of IPP along with Public utility. Ownership of projects is dependent on the financing structure. Generally, tenders are issued by Govt. utilities for which reverse bidding takes place for transparent discovery of solar tariff under Section 63 of Electricity Act 2003. Plain vanilla solar tenders are typically of two types predominantly, viz. solar park (BOOT) and Non-Solar Park (BOO models).

3.3.1 Build Own Operate (BOO) Model

In the BOO Model, the project land and complete responsibility of connectivity inter alia lies with the Project Developer. Complete ownership of project by IPP Developer / PPP Consortium.

Figure 14: Project Overview for BOO Model



Key Highlights:

⁹ Source: Deloitte Research

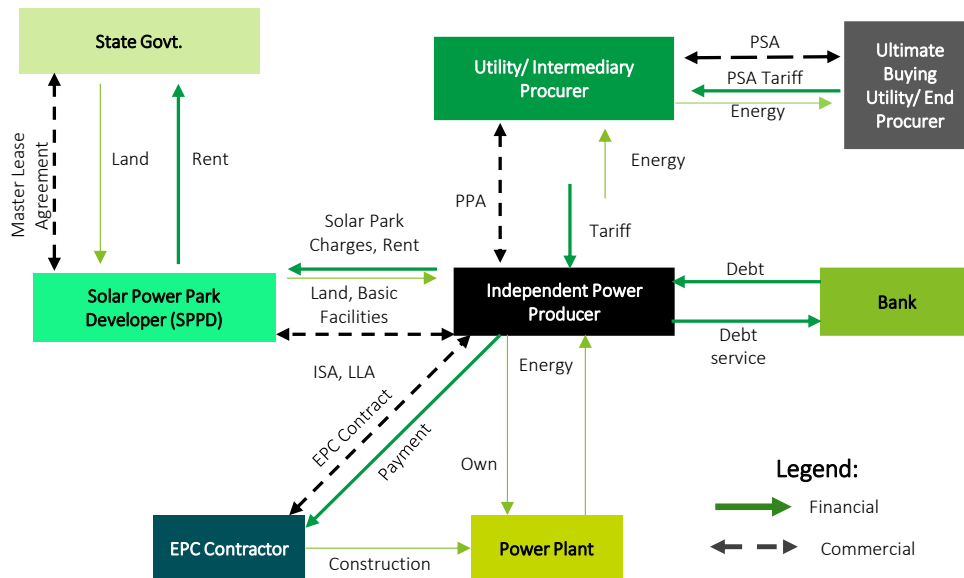
Aspect	Details
Key Features	- Suitable for large grid connected solar projects – can be within a solar park or independently developed - Based on BOOM or BOOT Model
Financing	- Equity by Private developers - Project specific debt from commercial banks/ FIs - Solar Park (wherever applicable) developed through FI funding + Government equity
Contractual Arrangement	- PPA signed with state distribution utility/ open access consumer or third party - ISA/ land lease agreement with solar park developer (if applicable) - Transmission Service Agreement with the transmission utility (State/ Central)
Commercial	Tariff based competitive bidding (TBCB) for procurement by utilities/ government agencies
Pros and Cons	Benefits - Cost optimization through TBCB route - Participation of Private players with niche expertise Challenges - Delays in case of dependence on other agencies for clearances

Details of BOO based solar projects implemented under SECI scheme is provided as Annexure 5.1

3.3.2 Build Own Operate Transfer (BOOT) – Solar Park Model

In case of solar park tenders, developers establish solar projects in pre-identified lands with clear demarcations; wherein responsibility of land and connectivity lies with Solar Park Implementing Agency (Public / JV with State Govt.) (SPIA). SPIA signs Master Lease Agreement with State Govt. and in turn signs Sub-Lease Deeds with Developer as well as Implementation Support Agreement (ISA). The Developer signs PPA with Intermediary Procurer/ Procurer and Intermediary Procurer in turn signs PSA with Ultimate Buying Entity. Since land is govt. owned, post PPA term, assets are transferred by Developer to Govt. entity/PPA party.

Figure 15: Project Overview of BOOT model (Solar Park)



Key Highlights:

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Aspect	Details
Key Features	• Land, solar park infrastructure along with off take infrastructure from PSS to GSS provided by Solar Park Implementing Agency (Public/JV) • Project Construction, connectivity upto PSS and O&M by Private Developer • After 25 years, asset transferred from Private to SPIA (Public/ JV)
Pros	• Ready land and basic infrastructure at the outset • Shorter turnaround time for project execution
Cons	• Implementing the model can be complex as it involves numerous transactions

State-wise details of solar parks as on Dec 2022 as Annexure **5.2**

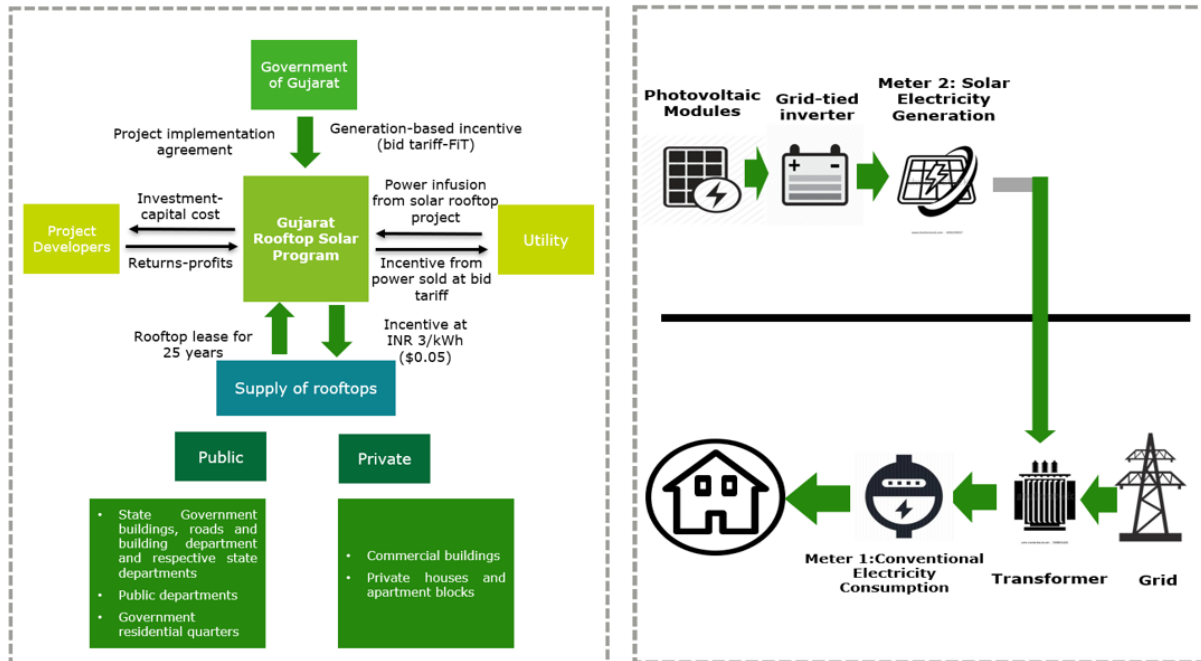
3.4 Distributed/Rooftop Solar Models

The Rooftop Solar segment also presents two major business models viz Gross Metered and Net Metered Models. These can be both utility / privately or PPP (Public Private Partnership) based projects. The model structure along with their live cases have been enumerated below:

3.4.1 PPP based Gross Metered Model – Example from Gujarat

In a gross metering model, entire generation of solar rooftop project is fed into the grid and sold to utility at the DISCOM offered rate. Rooftop owner does not consume the generated solar energy directly. A PPP based gross-metered Rooftop solar model adopted in Gujarat is presented below:

Figure 16: PPP based Gross-metered Solar rooftop model

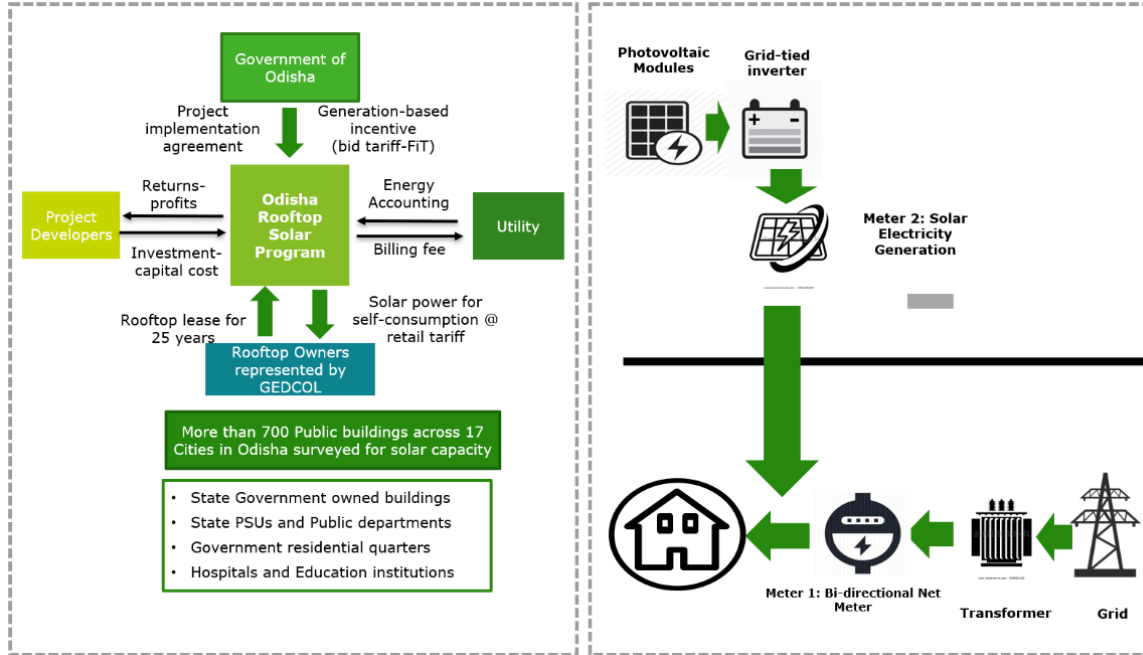


Private developer was selected through tender process. To mitigate the risk of rooftop availability, the government committed to lease public building rooftops to meet at least 80 percent of total capacity. A project development and management unit was designated to undertake a detailed technical, commercial, and regulatory assessment for the project.

3.4.2 PPP based Net Metered Model – Example from Odisha

In a net metering model, a bi-directional meter is installed along with the Rooftop solar project. The rooftop owner can consume the power generated from the Rooftop plant. In a net metered arrangement, excess generated power from solar plant can be fed into the grid through the net meter and during non-solar hours, power would be sourced from the grid again through the same meter. The net energy exported or imported would be factored in while utility would bill the consumer. PPP also potentially finds application here as is illustrated through the PPP based net metering project adopted in Odisha as presented below.

Figure 17: PPP-based Net metered Rooftop Solar Model



Government of Odisha partnered with the private sector to finance and set up solar panels on almost 200 buildings in the cities of Bhubaneswar and Cuttack with a minimum capacity of 4 MW. Transaction advisory was appointed by State-owned Green Energy Development Corporation of Odisha Limited, to assist in reviewing, structuring and implementing the project. A “Build Own Operate” (BOO) model was adopted here wherein, the winning bidder will set up solar panels across 200 Government and public sector buildings, with a minimum 4MW of installed capacity in Bhubaneswar and Cuttack. The lease agreement for Phase I of the project (200 KW) has been signed, and the design approved by the Independent Engineer.

Each business model presents unique advantages and challenges. The Public Sector Owned Utility Scale Solar Model could be beneficial for building local expertise, but the government's limited experience in managing large solar projects may pose challenges. The CPSU Scheme simplifies transaction structuring but requires careful consideration towards achieving the benchmarked tariffs. Private-owned models (BOO and BOOT) offer cost optimization and attract private players, but they may face delays and complexities in certain situations. Assam needs to carefully evaluate these models based on its specific requirements, resource availability, and administrative capabilities to choose the most suitable approach for sustainable solar project implementation. To identify suitable business model for implementation or transition between one model to another, it is also important to address some of the key challenges and learnings from solar project development in the state, as discussed in the previous section.

4 Way Forward

The State of Assam has witnessed a Compound Annual Growth Rate (CAGR) of ~10% in aggregate State GDP in the last 5 years (FY17 to FY22)¹ which is more than the National and NER growth rates. The State is a focus area state for the Govt. of India due to its strategic location and access point to the rest of NER. Backed by the economic growth, the power demand in Assam has been increasing at a CAGR of ~7.8% since FY16² and is expected to grow at a CAGR of 6.4% during the next decade till FY32.

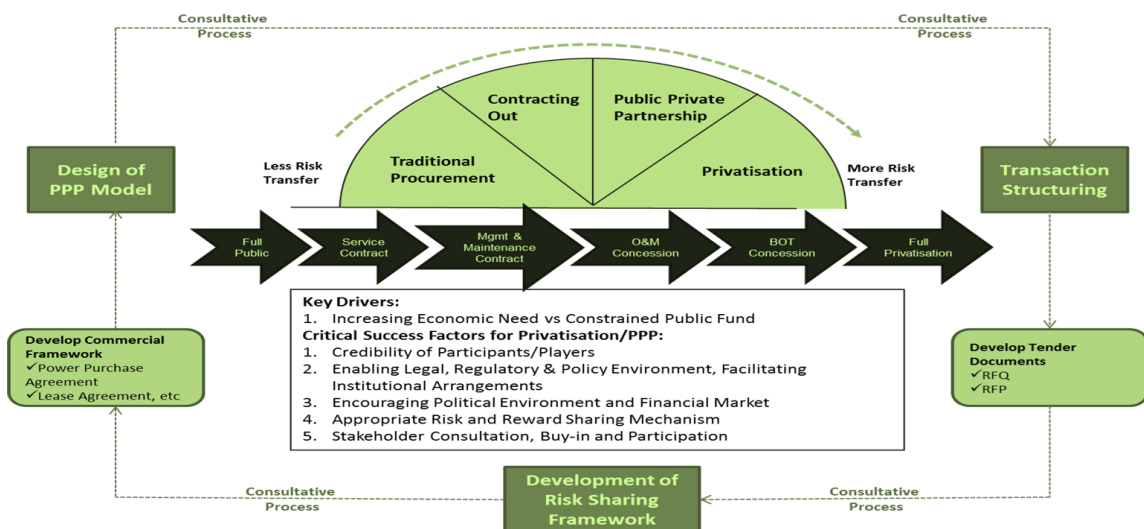
Assam's own electricity supply portfolio which has historically been dominated by Gas and Hydro based power plants are facing challenges ranging from wide price fluctuations, availability of gas as well as environment and gestation challenges. The State's APPC of INR ~5.4 per kWh in FY23 makes it one of the highest in the region.

The current energy landscape in Assam faces significant challenges, especially with the introduction of the new Renewable Power Obligation (RPO) mandated by the Ministry of Power (MoP). According to the proposed trajectory, Assam must strive for approximately 4200 MW of solar capacity by 2032 to fulfill its RPO. Despite possessing substantial potential for solar projects, the state is resorting to purchasing green power from short-term markets, even during solar hours, due to rising demand and inter-state transmission capacity constraints.

Given the limited resource potential for coal, wind, hydro, small hydro, and biomass in the state, solar emerges as the primary resource with a significant capacity of 13.7 GW for power supply expansion. However, Assam has faced challenges in implementing Independent Power Producer (IPP) and Public-Private Partnership (PPP) projects. Issues such as difficult terrain, high transportation costs, and power evacuation challenges have deterred private players, resulting in the realization of only a modest 70 MW capacity at a single location.

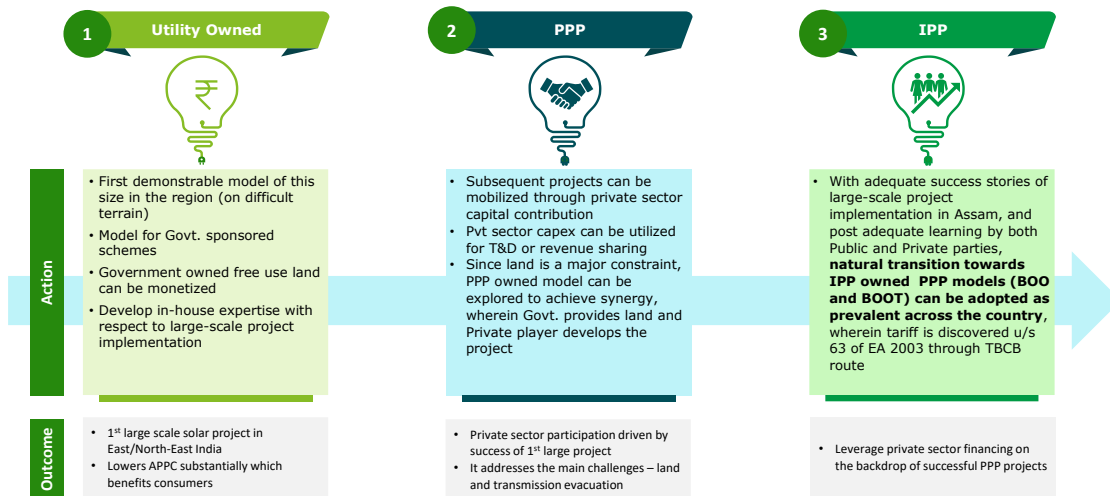
In response to these challenges, it becomes imperative to expedite the establishment of renewable energy projects in Assam, particularly at a large scale, to meet the surging demand and fulfill RPO requirements. The state utilities' support is crucial in initiating successful demonstrable large-scale projects. While private participation can be incorporated, a utility-led capital expenditure (capex) model with initial public utility responsibilities is envisioned. This strategic approach aims to address existing implementation challenges, such as land and evacuation issues, and lay the foundation for a gradual transition to PPP and eventually IPP models in the state.

Figure 18: Development of Implementation models for solar in a developing market



The phased approach, beginning with a utility-led capex model, ensures that the state utilities play a pivotal role in the early stages, taking on significant responsibilities. This involvement not only helps overcome initial hurdles but also establishes a framework for future large-scale projects and the achievement of market maturity in the state. As major capacity projects under the utility-led model prove successful, private participation can be gradually increased, leading to a natural evolution towards more private sector-led models. This transition allows for the gradual transfer of risks from the public to private players, fostering a sustainable and progressive development trajectory for renewable energy projects in Assam. Overall, this strategic and phased approach aims to create a conducive environment for the successful execution of large-scale projects, paving the way for Assam to meet its renewable energy targets and contribute substantially to the country's clean energy goals.

Figure 19: Impact of Various Implementation Models



5 Annexure

5.1 Annexure - Details of BOO based solar projects under SECI bidding

Table 6: List of Commissioned SECI Projects under BOO Model¹⁰

Sr. No.	Name of Project Developer	Project Capacity (MW)	Project Location (State)	Actual Commissioning Date	Total capacity commissioned as on date (MW)
Solar Tenders under SBG					
2000 MW ISTS Connected Solar PV Projects (ISTS-I)					
1	Mahindra Renewables Private Limited	250	Vill-Bhiv Ji ka Gaon, Tehsil-Bap, Dist.- Jodhpur	14.05.2021 (100 MW) 07.06.2021 (60 MW) 23.07.2021 (50 MW) 17.08.2021 (40 MW)	250
2	Kilaj Solar (Maharashtra) Pvt. Ltd	50	Vill:, Nure Ki Bhurj/Sihra, Dist: Jodhpur	17.Apr.2020	50
5	Azure Power Forty-Three Private Limited	300	Village Jagdev wala and Daudsar, Tehsil Bikaner, District Bikaner	15.12.2020 (150 MW) 30.01.2021 (100 MW) 08.02.2021 (50 MW)	300
6	Azure Power Forty-Three Private Limited	300	Village Jagdev wala and Daudsar, Tehsil Bikaner, District Bikaner	25.06.2021 (50 MW) 08.08.2021 (50 MW) 22.09.2021 (100 MW) 31.12.2021 (100 MW)	300
7	Clean Solar Power (Jodhpur) Pvt. Ltd	250	Vill-Noore Bhurj, Tehsil-Bap-Dist.-Jodhpur	14.03.2022 (115 MW) 21.04.2022 (135 MW)	250
750 MW Grid Connected in Rajasthan Tranche-I (non-solar park)					
2	Fortum Solar Plus Pvt. Ltd.	250	Vill: Bainsada , Tehsil: Pokhran, Dist: Jaisalmer, State Rajasthan	19.03.2021 (150 MW) 01.05.2021 (100 MW)	250
3	ReNew Solar Energy (Jharkhand Five) Pvt. Ltd	110	Vill: Bhainsara, Tehsil: Fatehgarh, Dist: Jaisalmer, State Rajasthan	04.02.2021 (50 MW) 11.03.2021 (60 MW)	110
4	Sitara Solar Energy Pvt. Ltd.	100	Vill: Kanasar, Tehsil: Bap, Dist: Jodhpur, State Rajasthan	20.May.2021	100
5	Palimarwar Solar House Pvt. Ltd.	20	Vill: Hariyasar, Tehsil: Pokhran, Dist: Jaisalmer, State Rajasthan	19.04.2021 (20 MW)	20
6	Palimarwar Solar House Pvt. Ltd.	20	Vill: Hariyasar, Tehsil: Pokhran, Dist: Jaisalmer, State Rajasthan	19.04.2021 (20 MW)	20
1200 MW ISTS Connected Projects (ISTS-III)					
1	SBSR Power Cleantech Eleven Pvt. Ltd.	300	Vill: Lalsar & Hapasara, Tehsil: Bikaner, Dist: Bikaner, State Rajasthan	15.08.2021 (50 MW) 04.04.2022 (50 MW) 11.04.2022 (50 MW)	150
2	ReNew Sun Waves Pvt. Ltd.	300	Vill: Mandhopura, Tehsil: Fatehgarh, Dist: Jaisalmer, State Rajasthan	13.08.2021 (150 MW) 17.08.2021 (50 MW)	300

¹⁰ Source: SECI

Sr. No.	Name of Project Developer	Project Capacity (MW)	Project Location (State)	Actual Commissioning Date	Total capacity commissioned as on date (MW)
Solar Tenders under SBG					
				04.09.2021 (50 MW) 04.10.2021 (50 MW)	
3	Eden Renewable Cite Pvt. Ltd.	300	Vill:Lakhasar, Tehsil: Pokhran, Dist: Jaisalmer, State Rajasthan	13.08.2021 (300 MW)	300
4	Azure Power Forty One Pvt. Ltd.	300	Vill: Noore Ki Bhurj, Khakhuri, Dedasari, Kushla Ram ki Basti, Ismail ki Dhani, Tehsil: Bap, Dist: Jodhpur, Rajasthan	12.10.2021 (50 MW) 30.10.2021 (50 MW) 30.11.2021 (50MW) 24.12.2021 (50 MW) 31.01.2022 (50 MW) 07.03.2022 (50 MW)	300
1200 MW ISTS Connected Projects (ISTS-IV)					
1	Ayana Renewable Power One Pvt. Ltd.	300	Vill: Khichiya, Bikaner Dist., Rajasthan	25.11.2021 (100 MW) 07.12.2021 (50 MW) 22.12.2021 (150 MW)	300
2	Renew Solar Energy (Jharkhand Three) Pvt. Ltd.	300	Vil: Mandhopura, Tehsil: Fatehgarh, Dist Jaisalmer, Rajasthan	02.09.2021 (150 MW) 27.09.2021 (50 MW) 04.10.2021 (50 MW) 09.12.2021 (50 MW)	300
3	Azure Power Maple Pvt. Ltd.	300	Vill: Sonanda, Shekhar, Bandhari & Kesarapura Teshsil: Bap, Dist. Jodhpur, Rajasthan	14.02.2022 (53 MW) 30.03.2022 (204 MW)	300
4	Mega SuryaUrja Pvt. Ltd.	250	Jodhpur Dist., Rajasthan	20.05.2022 (175 MW)	250
750 MW Grid Connected in Rajasthan Tranche-II (non-solar park)					
1	NTPC Limited	160	Vill: Jetsar, Dist. Sri Ganga Nagar, Rajasthan	22.10.2021 (80 MW) 25.03.2022 (80 MW)	160
2	Mahindra Susten Pvt. Ltd.	200	Bikaner Dist. Rajasthan	14.10.2021 (200 MW)	200
3	Clean Solar Power (BHAINSADA) Pvt. Ltd.	250	Vill: Bhainsada, Amarsar/Balwantgarh, Tehsil: Bhaniyana, Dist: Jaisalmer, Rajasthan	05.04.2022 (150 MW) 05.05.2022 (100 MW)	250
1200 MW ISTS Connected Projects (ISTS-V)					
1	GRT Jewellers (India) Pvt. Ltd.	150	Taluka: Ettayapuram, Kayathar, District: Thoothukkudi, Tamil Nadu	30.03.2022 (50 MW) 13.04.2022 (50 MW)	150
1200 MW ISTS Connected Projects (ISTS-VI)					
1	Avaada Sustainable RJProject Private Limited	300	Village: Noorsar, Tehsil: Bikaner in Bikaner, Rajasthan	14.01.2022 (100 MW) 07.02.2022 (50 MW) 12.04.2022 (50 MW) 11.05.2022 (100 MW)	300
2	Masaya Solar Energy Private Limited	300	Khandwa, Madhya Pradesh		150
3	ReNew Solar Urja Private Limited	300	Village Mandhopura, Tehsil Fatehgarh, District Jaisalmer, State Rajasthan	01.12.2021 (150 MW) 07.12.2021 (50 MW)	300

Sr. No.	Name of Project Developer	Project Capacity (MW)	Project Location (State)	Actual Commissioning Date	Total capacity commissioned as on date (MW)
Solar Tenders under SBG					
				13.12.2021 (50 MW) 16.12.2021 (50 MW)	
2000 MW ISTS Connected Projects (Tranche-IX)					
1	Thar Surya 1 Pvt. Ltd.	300	Vill: Dholera 1, Jamsar, Dist: Bikaner, Rajasthan	23.03.2022 (54 MW) 23.06.2022 (54 MW) 09.08.2022 (54 MW) 11.10.2022 (74 MW) 24.11.2022 (64 MW)	300

Note: ~5.6 GW projects commissioned as on August 2023 (excl. CPSU projects under SECI)

5.2 Annexure – Details of Solar Park (BOOT) based solar projects across states

Table 7: State Wise Details of Solar Parks¹¹

#	State	Name of Solar Park	Approved Capacity (MW)	Status
1	Andhra Pradesh	Ananthapuramu-I Solar Park	1400	Completed
2		Kurnool Solar Park	1000	Completed
3		Kadapa Solar Park	1000	
4		Anathapuramu-II Solar Park	500	Completed
5		Solar-Wind Hybrid Solar Park	200	
6	Chhattisgarh	Rajnandgaon Solar Park	100	
7	Gujarat	Radhanesda Solar Park	700	
8		Dholera Ph-I Solar Park	1000	
9		NTPC RE Park	4750	
10		GSECL RE Park	3325	
11		GIPCL RE Park Ph-I	600	
12		GIPCL RE Park-Ph-II	1200	
13		GIPCL RE Park-Ph-III	450	
14	Himachal Pradesh	Kaza Solar Park	880	
15		Kinnaur Solar Park	400	
16	Jharkhand	Floating Solar Park	100	
17		Deogarh Solar Park	20	
18		Palamu Solar Park	20	
19		Garwha Solar Park	20	
20		Simdega Solar Park	20	
21	Jharkhand & West Bengal	DVC Floating Solar Park Ph-I	755	
22		DVC Floating Solar Park Ph-II	234	
23	Karnataka	Pavagada Solar Park	2000	Completed
24		Kalburgi Solar Park	500	
25	Kerala	Kasargod Solar Park	105	
26		Floating Solar Park	50	
27	Madhya Pradesh	Rewa Solar Park	750	Completed
28		Mandsaur Solar Park	250	Completed
29		Neemuch Solar Park	500	
30		Agar Solar Park	550	
31		Shajapur Solar Park	450	
32		Chhatarpur Solar Park	950	
33		Omkareshwar Floating Solar Park	600	
34		Morena Solar Park	1400	
35		Barethi Solar Park	630	

¹¹ Source: Lok Sabha Unstarred Question No. 440 for 08.12.2022

#	State	Name of Solar Park	Approved Capacity (MW)	Status
36	Maharashtra	Sai Guru Solar Park	500	
37		Dondaicha Solar Park	250	
38	Mizoram	Vankal Solar Park	20	
39	Odisha	Solar park by NHPC	40	
40		Floating Solar Park Ph-I	100	
41		Floating Solar Park Ph-II	200	
42	Rajasthan	Bhadla Phase-II Solar Park	680	Completed
43		Bhadla III Solar Park	1000	Completed
44		Bhadla IV Solar Park	500	Completed
45		Phalodi-Pokaran Solar Park	750	
46		Fatehgarh Phase 1B Solar Park	421	
47		Nokh Solar Park	925	
48		Pugal Solar Park Phase-I	1000	
49		Pugal Solar Park Phase-II	450	
50		RVUN Solar Park	1310	
51	Uttar Pradesh	Solar Park in UP	365	
52		Jalaun Solar Park	1200	
53		Kalpi Solar Park	65	
54		Mirzapur Solar Park	100	
55		Lalitpur Solar Park	600	
56		Jhansi Solar Park	600	
57		Chitrakoot Solar Park	800	
			39285	

~39.28 GW approval for solar parks have been accorded out of which, ~8 GW solar parks have been completed.

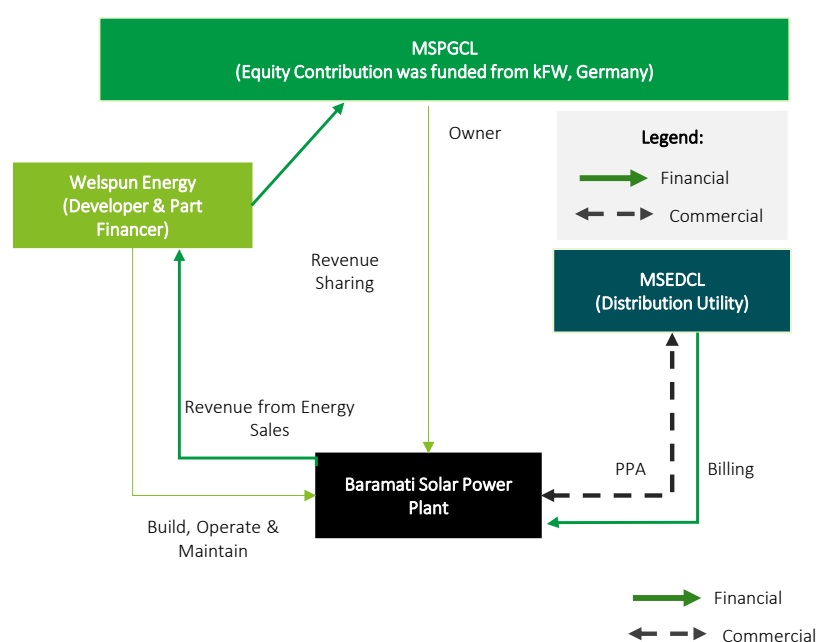
5.3 Annexure - Utility Owned Model – Private Participation - Mahagenco Case Study:

A Case Study of Utility Owned Model with Private Participation has been enumerated below:

Project Details:

- Maharashtra State Power Generation Company Limited (MSPGCL / Mahagenco) owns a 50 MW Grid Connected Solar Project in Baramati Maharashtra
- The project was a unique PPP model in India as it involved revenue share with the project owner for 25 years (linked to PPA)
- Welspun India developed the project on a Build, Operate and Maintain basis with MSPGCL

Figure 20: Project Overview



Key Highlights:

Table 8: Key Highlights of Mahagenco Project

Aspect	Details
Project Funding	Jointly funded by MSPGCL (Owner) & Welspun (Developer)
Mode of selection of Developer	The successful bidder was selected based on the net present value of the highest revenue share offered to MSPGCL over the life of the project.
Power Purchase Agreement	PPA with MSEDC at Tariff Rate discovered via Bidding process
Revenue Sharing	- The bidder with the highest revenue sharing with MSPGCL was offered the contract. - Thus Welspun offered 38.75% revenue sharing and won the project - The revenue share was offered assuming CUF of 19%

6 End of Document



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